MILITARY SPECIFICATION SHEET

BATTERY, NON-RECHARGEABLE, LITHIUM SULFUR DIOXIDE BA-5588/U

This specification sheet is approved for use within Laboratory Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the lithium sulfur dioxide non-rechargeable battery described herein shall consist of this document and the latest issue of MIL-B-49430(ER).

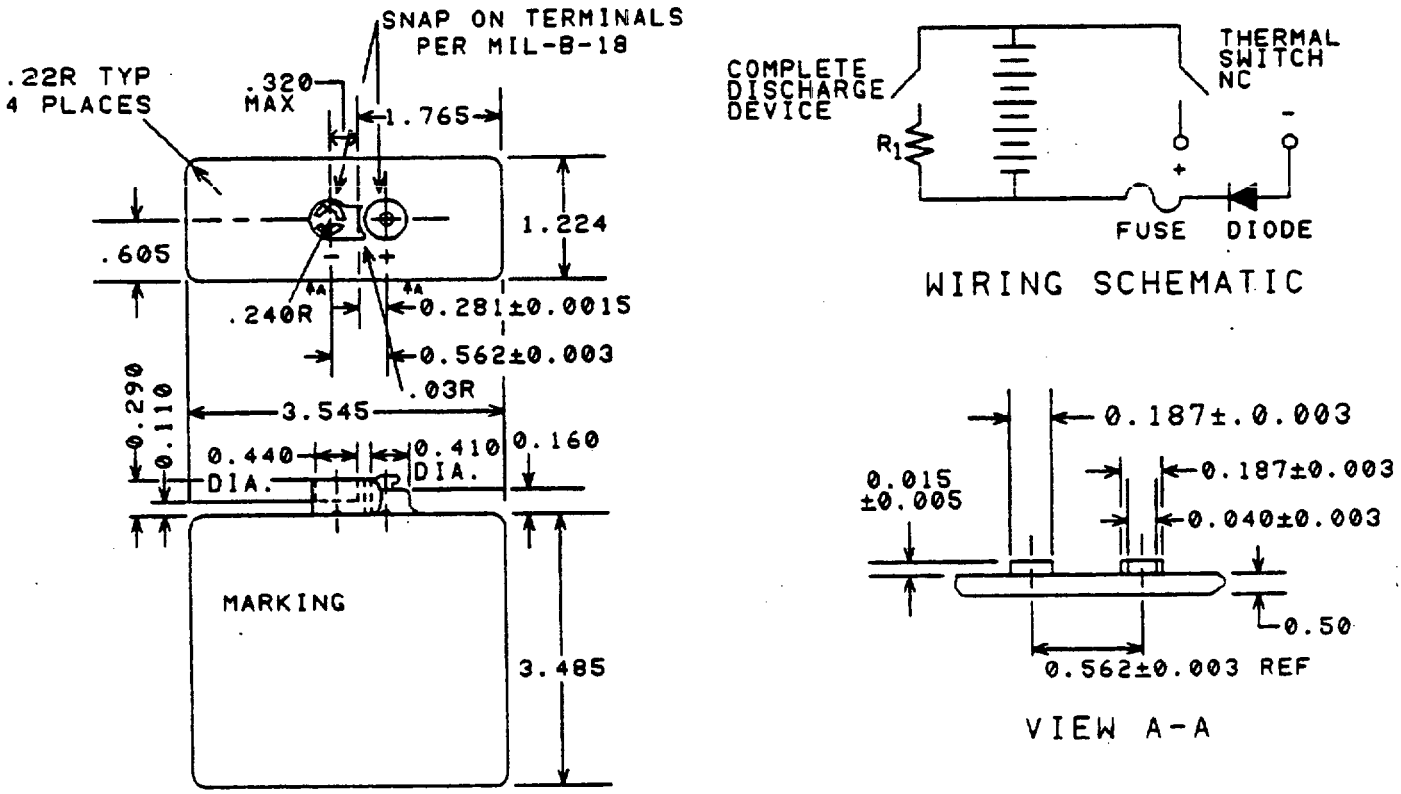


Figure 1. Battery dimensions and wiring schematic.

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NOTES:

1. All dimension are in inches.

2. Tolerances, unless otherwise specified, are ±.015 inch.

3. The battery connector shall be snap-on terminals per MIL-B-18 with protective shield (negative terminal) as shown in Figure 1.

4. Socket wiring schematic shall be printed or embossed on the jacket near the connector on the terminal face of the battery.

5. R_1 value shall be between 300 ohms and 430 ohms and have a power rating of 1.0 watt.

REQUIREMENTS:
Dimensions and configuration: See Figure 1.

Maximum voltage: 15.25 volts.

Minimum voltage (for capacity): 11.00 volts.

The battery shall contain a 1 ampere non-replaceable slow blow fuse wired in series with cells in the negative leg and a thermal switch or fuse.

The battery shall contain a diode wired in series. See paragraph 3.5.5.2 of the Basic Specification.

Terminals: Snap-on type (see Figure 1).

Weight (maximum): 0.65 pound (295 grams).

Capacity tests: When the battery is tested in accordance with the methods of examination and tests of this specification, the minimum capacity test requirements shall be as specified below:

Capacity Test	Battery Capacity Requirements in Hours to 11.00 Volts
I	3.6
L	1.7
H	3.6
HT	3.1
LT	1.6
T	3.3
D	3.3
ILR	1000

Initial voltage delay: During the I, H, HT, T, D, and ILR discharge tests, initial voltages below the 11.00 volt minimum voltage cannot exceed a 10 second duration. During L and LT discharge tests initial voltage below 10.00 volts cannot exceed a 10 second duration.

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High temperature switch: A high temperature switch or thermal fuse shall be employed internal to the battery in the series circuit. As described in paragraphs 3.5.5 and 3.5.5.1 of the Basic Specification.

Complete discharge device: This device shall be inserted in the battery as shown on the circuit schematic and shall have the characteristics as described in para. 3.5.5.3 of the basic specification.

METHODS OF EXAMINATION AND TESTS

Capacity Tests:

(1) Storage: Details on storage conditions for all specified capacity tests are described in the basic specification.(See 4.8.9)

(2) Discharge:

Cells: Discharge at 0.8 amps constant current at 70\5^F for a minimum of 3.1 hours to 2.2 volt cutoff. Continue discharge to zero volts, then discharge at 0.4 amperes constant current for 3.1 hours into voltage reversal.

Batteries: For all capacity tests except the ILR, the batteries shall be discharged at 0.8 amperes constant current to zero volts. For the I capacity test the batteries shall be forced discharged at 0.4 amperes constant current for 5.6 hours immediately after zero volts is obtained. There shall be no venting, fire or explosion when the batteries are force discharged. For the ILR test a fixed resistance for 5,600 ohms shall be used to a cut-off voltage of 11.00 volts or 1000 hours.

Closed circuit voltage: Load resistance of 25 ohms shall be used between battery terminals. Voltage shall be above 11.00 volts within 10 seconds.

Revision letters are not used to denote changes due to the extensiveness of the changes.

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